

Hostel Management System

DATABASE MANAGEMENT AND WAREHOUSING

DR. RYAN LABRIE

CHARMY RAJ | SPU ID:900420893

Table of Contents

Business Justification for Hostel Management System.....	1
Application Section.....	2
ER-Diagram.....	2
Relational Schema	3
User Interface	5
Forms	6
Reports.....	10
Conclusion.....	12
What went well?	12
What did not go well?	13
What would you do differently next time?	13

Business Justification for Hostel Management System

Managing a hostel efficiently is important for providing a smooth and comfortable living experience for students and staff. As the number of hostel residents increases, using paper records or outdated systems makes managing rooms, payments, complaints, and daily activities difficult. To solve these issues, we need a Hostel Management Database System that will simplify operations and improve management efficiency.

The new system will help with the following challenges:

1. **Better Data Management:** Right now, student details, room assignments, fees, complaints, and mess records are stored separately, making it hard to find and manage information. This database will store all data in one place, making access easier and improving decision-making.
2. **Room Allocation System:** Assigning rooms manually can lead to mistakes, such as double booking or incorrect records. The system will automate room assignments, show available rooms in real-time, and keep accurate records, making the process easier for administrators.
3. **Fee Payment Tracking:** Keeping track of student fee payments and overdue balances can be challenging. The system will automatically update payment records, send reminders for due payments, and ensure that financial records are always up to date.

4. **Handling Complaints Efficiently:** Students often face issues related to hostel facilities. The system will allow students to file complaints online, track their status, and help administrators resolve them quickly.
5. **Mess Monitoring:** Tracking mess usage manually can lead to errors. This system will help manage meal planning and reduce food wastage.
6. **Status Management for Better Organization:** The database will have different status tables to manage room availability, fee payments, complaint progress, and mess participation, ensuring everything is well-organized.
7. **Improved Decision-Making:** The system will generate reports on hostel occupancy, payment collections, complaint resolutions, and other important areas. These insights will help management make better plans and decisions.

Benefits:

- **Smoother Operations** – Automates and simplifies data management.
- **Better Student Experience** – Ensures efficient room allotments, quick complaint handling, and easy meal planning.
- **Transparent Fee Management** – Keeps accurate records of payments and dues.
- **Improved Resource Management** – Tracks occupancy and mess attendance in real-time.
- **Scalability** – Can easily handle future hostel expansions and changing requirements.

The database system will be built using Microsoft Access, which is easy to use and offers strong data management features. It will consist of several interconnected tables, such as Students, Rooms, Fees, Complaints, Mess, and Status Tables, to store and manage data effectively. This digital system will help hostel management run smoothly, ensure transparency, and improve the living experience for students.

Application Section

ER-Diagram

I have provided a detailed description of the Entity-Relationship (ER) diagram, outlining the entities (tables) I have modeled, their attributes, and the relationships between them. The entities I have identified, such as Student, Rooms, RoomAllotments, Complaints, Mess, Fees, and MealTypes, accurately reflect the core components of a hostel management system.

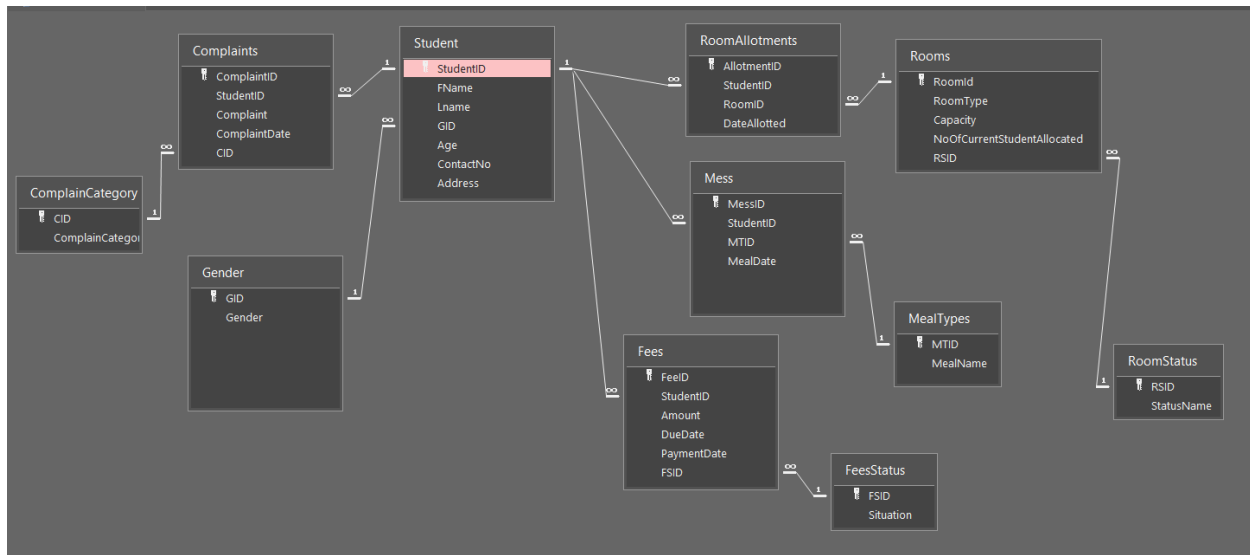


Fig 1 ER-Diagram of Hostel Management System

- Student is the central entity, connected to complaints, fees, room allotments, and mess.
- Rooms are allocated through RoomAllotments and have statuses.
- Meals are recorded in the Mess entity, with types stored in MealTypes.
- Fees management is handled through the Fees entity, linked to FeeStatus.

This ERD represents a hostel management system that tracks students, their accommodations, complaints, meals, and payments efficiently.

Relational Schema

1. Student Table

Schema: Student(StudentID, FName, LName, GID, Age, ContactNo, Address)

- Primary Key (PK): StudentID
- Foreign Key (FK): GID → Gender(GID)

2. Gender Table

Schema: Gender(GID, Gender)

- Primary Key (PK): GID

3. RoomAllotments Table

Schema: RoomAllotments(AllotmentID, StudentID, RoomID, DateAllotted)

- Primary Key (PK): AllotmentID
- Foreign Keys (FK):
 - o StudentID → Student(StudentID)
 - o RoomID → Rooms(RoomID)

4. Rooms Table

Schema: Rooms(RoomID, RoomType, Capacity, NoOfCurrentStudentAllocated, RSID)

- Primary Key (PK): RoomID
- Foreign Key (FK): RSID → RoomStatus(RSID)

5. RoomStatus Table

Schema: RoomStatus(RSID, StatusName)

- Primary Key (PK): RSID

6. Mess Table

Schema: Mess(MessID, StudentID, MTID, MealDate)

- Primary Key (PK): MessID
- Foreign Keys (FK):
 - o StudentID → Student(StudentID)
 - o MTID → MealTypes(MTID)

7. MealTypes Table

Schema: MealTypes(MTID, MealName)

- Primary Key (PK): MTID

8. Fees Table

Schema: Fees(FeeID, StudentID, Amount, DueDate, PaymentDate, FSID)

- Primary Key (PK): FeeID
- Foreign Keys (FK):
 - o StudentID → Student(StudentID)

- o FSID → FeesStatus(FSID)

9. FeesStatus Table

Schema: FeesStatus(FSID, Situation)

- Primary Key (PK): FSID

10. Complaints Table

Schema: Complaints(ComplaintID, StudentID, Complaint, ComplaintDate, CID)

- Primary Key (PK): ComplaintID
- Foreign Keys (FK):
 - o StudentID → Student(StudentID)
 - o CID → ComplainCategory(CID)

11. ComplainCategory Table

Schema: ComplainCategory(CID, ComplainCategory)

- Primary Key (PK): CID

User Interface

User Interface: I have developed a comprehensive user interface consisting of multiple forms and reports to meet the business needs of hostel management. The forms I have created, such as the Student Registration Form, Room Allocation Form, and Fee Payment Form, allow for efficient data entry, retrieval, and management of student, room, and fee details. Validation rules have been implemented for fields like contact numbers and email addresses to ensure data integrity and improve user experience. Additionally, I have incorporated subforms and calculation fields to enhance system functionality and usability. For instance, the Room Allocation Form includes a subform displaying current room availability and assigned students, while the Fee Payment Form calculates total due amounts and tracks pending payments. The reports I have developed, such as the Pending Complaints Report and Fee Due Report, provide valuable insights into hostel operations and facilitate data-driven decision-making.

Populated & Functioning Database: I have successfully populated the database, enforcing domain, entity, and referential integrity constraints to maintain data consistency and accuracy. The inclusion of features like a main management dashboard, with buttons and macros to navigate between different forms and reports, enhances the overall user experience and operational efficiency. Furthermore, I have

incorporated unique and engaging features, such as alerts for overdue fees, automatic status updates for complaints, and a digital room allocation system, making the hostel management process more efficient and seamless. This system ensures transparency, automation, and ease of access, creating a well-organized and user-friendly hostel management experience.

Overall, the hostel management database system I have developed demonstrates a comprehensive understanding of database design principles, effective implementation of functional requirements, and a commitment to delivering a polished and integrated product. The system's capabilities in streamlining operations, improving student and staff interactions, and enabling data-driven decision-making position the hostel for better efficiency, transparency, and organization.

Forms

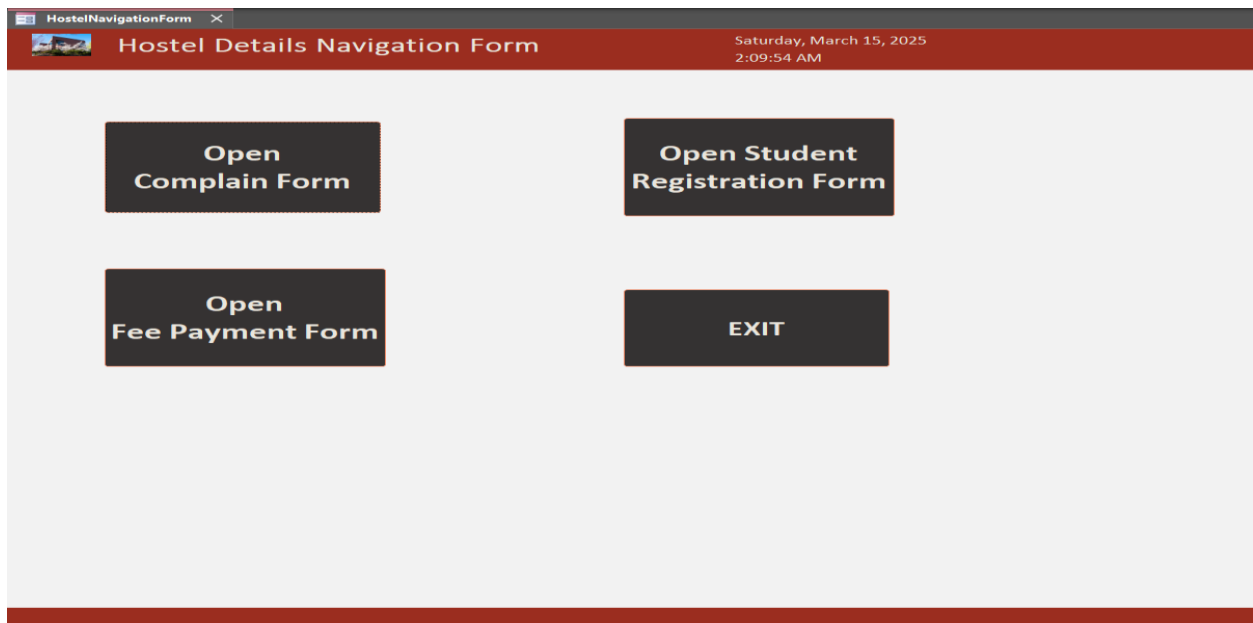
The image shows a screenshot of a web application window titled "HostelNavigationForm". The window has a red header bar with the text "Hostel Details Navigation Form" on the left and the date and time "Saturday, March 15, 2025 2:09:54 AM" on the right. Below the header, there are four dark blue buttons with white text arranged in a 2x2 grid. The buttons are labeled "Open Complain Form", "Open Student Registration Form", "Open Fee Payment Form", and "EXIT". The background of the form is light gray.

Fig 2 Navigation Form [Form-1]

This form is set as the default form, meaning it will automatically open whenever the database is launched. It serves as a navigation hub for the hostel management system, providing access to all essential forms.

Users can open specific forms by clicking the respective buttons:

- Complaint Form – To register complaints.
- Student Registration Form – For new student enrollment.
- Fee Payment Form – To manage fee transactions.

Additionally, clicking the Exit button will close the application. This streamlined interface ensures easy access to key hostel management functions.

Complain Form

[illegible]

Fig 3 Student Complain Form [Form-2]

This form consists of two subforms:

1. Pending Complaints – Displays complaints that have not yet been resolved.
2. Resolved Complaints – Lists complaints that have been addressed.

Each subform includes a button that allows users to print a report for the respective complaints, ensuring easy record-keeping and tracking of complaint statuses.

Fee Payment Form

Fee Payment Form

StudentID	10		
StudentName	Aarav		
Amount	10000		
DueDate	3/26/2025	PaymentDate	
Fees Status	Pending	Print Pending Fees Report	

Record: 1 of 19 | No Filter | Search

Fig 4 Fee Payment Form [Form-3]

- This form allows users to identify students with overdue and pending hostel fees.
- The fee status is managed using a drop-down (combo box), enabling easy selection and updating of payment statuses.
- This setup ensures efficient tracking and management of student fee records.

Student Registration Form

Student Listing/Registration Form

StudentID	1
First Name	Kavya
Last Name	Patel
Age	19
Gender	Male Female
ContactNo	2067862502
Address	15th street EW

Print Report

Fig 5 Student Registration Form [Form-4]

- In this form, a list box is used for selecting the gender attribute.
- Additionally, when clicking the "Print Report" button, the system generates a report specifically for the selected student, rather than displaying a report for all students.
- This functionality ensures a more personalized and efficient report generation process.

Reports

Pending Complaints Report

ComplaintID	First Name	Last Name	Complaint	ComplaintDate	ComplainCategory
3	Rahul	Verma	The food quality in the mess is not satisfactory.	3/13/2025	Pending
5	Ishita	Malhotra	There are frequent power outages in our block.	3/8/2025	Pending
7	Shoppy	Good	The washrooms are not cleaned regularly, causing an unhygienic environment.	3/8/2025	Pending
10	Leki	Dutt	The hostel Wi-Fi is too slow and disconnects frequently, making it difficult to study.	3/13/2025	Pending
12	Aditya	Sharma	The hostel laundry service is inefficient, and clothes are not returned on time.	3/4/2025	Pending
14	Rohan	Iyer	There are too many mosquitoes and cockroaches in the hostel rooms and mess area.	3/11/2025	Pending
15	Nia	Sharma	My roommate keeps bringing guests without informing me, making it uncomfortable.	3/10/2025	Pending
16	Juan	Shah	My bed and room are infested with bed bugs, causing discomfort and allergies.	3/8/2025	Pending
17	John	William	There is a shortage of clean drinking water in the hostel.	3/23/2025	Pending
20	Virat	Kholi	There is not enough cupboard or storage space for my belongings.	3/27/2025	Pending

Saturday, March 15, 2025

Page 1 of 1

Fig 6 Pending Complaints Report [Report- 1]

- This report provides a list of students with unresolved complaints.
- It is also linked to the complaint form under the "Pending Complaints" section.

Resolved Complaints Report

ComplaintID	First Name	Last Name	Complaint	ComplaintDate	ComplainCategory
1	Kavya	Nair	The water supply in the bathroom is inconsistent.	3/6/2025	Resolved
2	Aarav	Kapoor	The WiFi is too slow and disconnects frequently.	3/12/2025	Resolved
4	Stephen	Gems	The fan in my room is making a loud noise.	3/6/2025	Resolved
6	Kavya	Patel	The fan in my room is not working properly, and it's getting too hot.	3/4/2025	Resolved
8	Aarav	Patel	The food served in the mess is not fresh and lacks variety.	3/7/2025	Resolved
9	Vikram	Singh	There is an inconsistent water supply, especially in the mornings.	3/8/2025	Resolved
11	Rahul	Verma	Some students play loud music late at night, disrupting sleep and study time.	3/10/2025	Resolved
13	Sanya	Mehta	Strangers are frequently seen in hostel premises without proper checks.	3/19/2025	Resolved
18	Victoria	Adams	The mess timings are too strict, making it difficult for students with different schedules.	3/17/2025	Resolved
19	Sushi	Sharma	Frequent power cuts make it difficult to study and complete assignments on time.	3/28/2025	Resolved

Saturday, March 15, 2025

Page 1 of 1

Fig 7 Resolved Complaints Report [Report- 2]

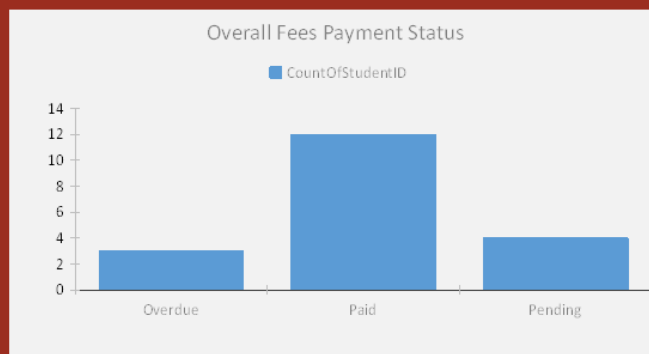
- This report provides a list of students with resolved complaints.
- It is also linked to the complaint form under the "Resolved Complaints" section.

Fees Payment Receipt

Student ID	First Name	Last Name	Amount	DueDate	Situation
10	Aarav	Kapoor	10000	3/26/2025	Pending
11	Aditya	Sharma	10000	3/22/2025	Pending
14	Juan	Shah	4000	3/31/2025	Pending
12	Nia	Sharma	7000	3/22/2025	Pending
9	Rohan	Iyer	15000	3/22/2025	Overdue
16	Victoria	Adams	3000	3/22/2025	Overdue
4	Vikram	Singh	5000	3/22/2025	Overdue

Grand Total

54000



Saturday, March 15, 2025

Page 1 of 1

Fig 8 Fees Payment Report [Report- 3]

- This report lists all students with overdue or pending hostel fees, along with the corresponding outstanding amounts.
- It also includes a grand total of the outstanding balance, along with the overall fees that have not been paid.

Student Hostel Details

StudentID	First Name	DateAllotted	RoomID	RoomType	MealDate	MealType
1	Kavya	1/15/2025	101	Single	3/10/2025	Breakfast

Saturday, March 15, 2025

Page 1 of 1

Fig 9 Student Hostel Detail Report [Report- 4]

- This report is linked to the student registration form and provides data for the selected student.
- To link to the student registration form, I was referential integrity formula.

Conclusion

What went well?

1. Clear understanding of project requirements and objectives: From the beginning, I had a strong grasp of the project's goals and the essential features needed for the hostel management database system. This allowed me to plan and execute the project effectively, ensuring all necessary components were addressed and implemented successfully. Effective requirement gathering and thorough analysis helped streamline the development process and ensured that all key functionalities were incorporated.
2. Successful implementation of key database features: I was able to implement the core functionalities of the system, including user-friendly forms for data entry, insightful reports for analysis, and validation rules to maintain data integrity. The addition of automation features, such as automatic complaint status updates and fee tracking, contributed to a more efficient and functional system. The use of lookup tables for standardizing data, ensuring referential integrity, and reducing redundancy significantly improved performance and data management.

3. Integration of unique and engaging elements: To enhance the user experience, I included features like an intuitive navigation dashboard, real-time message alerts, and interactive reports. These features improved the overall usability and professionalism of the system. The use of macros and buttons for easier navigation made the system more user-friendly, ensuring that both administrators and students could efficiently interact with the interface.
4. Strong validation and error handling mechanisms: To maintain data accuracy, I implemented input validation rules for fields. This significantly reduced errors and improved system reliability. The use of error messages and alerts helped users quickly identify and correct mistakes.
5. Scalability and flexibility: The database system was designed in a way that allows future expansion, such as adding new reports, integrating additional automation, or extending functionalities like mess management and attendance tracking.

What did not go well?

1. Underestimation of project complexity: Initially, I underestimated the complexity of integrating multiple automation features, which led to delays in troubleshooting and refining certain functionalities. Debugging macros and setting up correct relationships took longer than expected, impacting the timeline.
2. Challenges in handling large data volumes: While the database performed well with small datasets, performance issues arose when working with larger datasets, requiring additional optimizations. Queries and reports took longer to load, highlighting the need for better indexing and query structuring.
3. Limited experience with advanced database concepts: My limited knowledge of advanced queries and database optimization techniques restricted the implementation of more complex functionalities that could have further improved system performance. Stored procedures and triggers could have been used to automate certain tasks, but due to time constraints, these were not fully explored.
4. Graphical User Interface (GUI) design limitations: While the forms were functional, the overall aesthetic design of the user interface could have been more polished. A more visually appealing design with better UI/UX elements would have enhanced the user experience.

What would you do differently next time?

1. Focus on database optimization techniques: To prevent performance issues with large datasets, I would prioritize learning and implementing optimization techniques such as indexing, query optimization, and efficient data structuring.

Using partitioning and caching strategies could further improve system responsiveness.

2. Enhance automation and advanced database functionalities: I would dedicate more time to learning advanced database concepts such as stored procedures, triggers, and database indexing to improve automation within the system. This would allow for more dynamic data processing and reduce manual interventions.
3. Improve database design skills: In future projects, I would work on refining my database design skills to ensure better scalability, maintainability, and efficiency. Studying normalization techniques and implementing best practices in database architecture would help create a more robust system.
4. Allocate more time for GUI improvements: A visually appealing interface improves usability and engagement. Next time, I would focus on enhancing the layout, colors, fonts, and user experience design to make the system more intuitive and appealing to users.
5. Comprehensive testing and debugging: While I tested the system thoroughly, I would allocate more time for extensive testing with larger datasets and real-world scenarios to uncover potential performance bottlenecks and usability issues before deployment.